Peter Nguyen

(913) 912-9604 | peternguyen5555@gmail.com | linkedin.com/in/peter |

Education

Kansas State University Manhattan, KS
Master's of Science in Computer Engineering *July 2023 – Dec. 2024*

Kansas State University Manhattan, KS
Bachelor's of Science in Computer Engineering *Aug. 2019 – May 2023*

Experience

Graduate Research Assistant July 2023 – Dec. 2024
Kansas State University *Manhattan, KS*

- Conducted independent research on convolution and Fourier transforms in two-dimensional space
- Designed and implemented solutions on the Terasic DE1-SoC platform to support research objectives
- Coordinated weekly progress meetings with major advisor to ensure alignment with project goals and deliverables

Graduate Teaching Assistant Aug. 2023 – Dec. 2024
Kansas State University *Manhattan, KS*

- Assisted in supervising and guiding 16 students in lab assignments utilizing the Terasic DE1-SoC Board
- Collaborated with professor to design and refine laboratory exercises, enhancing hands-on learning experiences
- Provided targeted academic support through scheduled office hours, addressing student questions and troubleshooting technical issues

Projects

FFT Sobel Edge Detector Hardware Acclerator | *Terasic DE1 SoC* July 2023 – Nov 2024

- Utilized on-board SRAM to enable data communication between the FPGA and HPS
- Integrated an open-source FFT core to accelerate image edge detection processing
- Implemented an Intel External Bus to Avalon Bridge controller for shared SDRAM access by both FPGA and HPS
- Developed C programs to process 640×480 RGB webcam images into 128×128 grayscale format for FPGA-based processing

GPU Rendering | *Terasic DE2-115* May 2024 – July 2024

- Designed an SRAM controller to efficiently retrieve image data for rendering
- Developed FPGA modules to output image data from SRAM to VGA display
- Created a SPI slave module to interface with the Analog Discovery 2 Board for data acquisition

AES Encryption | *Terasic DE2-115* Mar. 2023 – May 2023

- Designed and implemented FPGA-based AES encryption and decryption modules
- Applied bit-shifting techniques to improve computational speed
- Developed a key generation module and S-Box lookup table for encryption and decryption operations
- Programmed display modules to present encoded and decoded outputs on seven-segment displays

LUP Decomposition | *Terasic DE1 SoC* Oct. 2022 – Dec. 2022

- Implemented LUP decomposition in FPGA using a Finite State Machine (FSM) design
- Integrated parallel I/O interfaces for data transfer between FPGA and HPS
- Leveraged Intel floating-point arithmetic IP cores for high speed computations

Technical Skills

Languages: SystemVerilog, Verilog, Python, C
Hardware: Altera Cyclone V SE SoC FPGA, Altera Cyclone IV FPGA, Analog Discovery 2 Board
Tools: Intel Quartus Prime, ModelSim, MATLAB, Linux, Signal Tap Logic Analyzer, WaveForms
Communication Protocols: SPI